ETALENESS OF A MORPHISM OF ALGEBRAIC STACKS TESTED THROUGH AN ETALE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Allowed standard prerequisites (editorial).

- The characterization of unramified morphisms used for one of the equivalences. Invoked at the start of the counted proof (source lines 8551-8553). Gives the equivalence of two of the listed conditions.
- The characterization of DM morphisms through surjective etale covers. Invoked inside the counted proof (source lines 8555-8559). Produces the algebraic spaces presenting the morphism.
- The statement that an unramified flat morphism of finite presentation of algebraic spaces is etale, together with the trivial implications from etaleness. Invoked at the end of the counted proof (source lines 8560-8566) as cross-chapter pointers. Finishes the equivalence for the presenting morphism.

Source reference. Original lemma lemma-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4746–4746. Navigation pointer only: the counted text refers to the Stacks item lemma-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-characterize-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8450–8450. Navigation pointer only: the counted text refers to the Stacks item lemma-characterize-unramified, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference spaces-morphisms-lemma-etale-flat: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8566–8566. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-flat, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-locally-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3672–3672. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-locally-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8317–8317. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-unramified, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-unramified-flat-lfp-etale: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1131–1131. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-unramified-flat-lfp-etale, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent*

(1) *f is étale, and*
(2) *f is locally of finite presentation, flat, and unramified,*
(3) *f is locally of finite presentation, flat, and its diagonal is étale.*

Proof. The equivalence of (2) and (3) follows immediately from Lemma lemma-characterize-unramified. Thus in each case the morphism f is DM. Then we can choose Then we can choose algebraic spaces U, V, a smooth surjective morphism $V \to \mathcal{Y}$ and a surjective étale morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ (Lemma lemma-DM). To finish the proof we have to show that $U \to V$ is étale if and only if it is locally of finite presentation, flat, and unramified. This follows from Morphisms of Spaces, Lemma spaces-morphisms-lemma-unramified-flat-lfp-etale (and the more trivial Morphisms of Spaces, Lemmas spaces-morphisms-lemma-etale-unramified, spaces-morphisms-lemma-etale-locally-finite-presentation, and spaces-morphisms-lemma-etale-flat). □